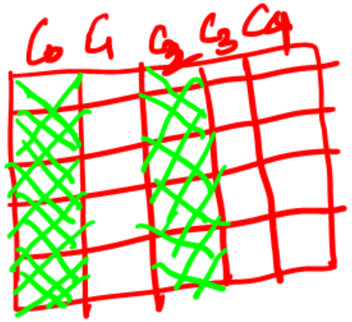
Project(Π) :→

Employee{Empid, Empname, salary, age, DOJ, Deptno.}.



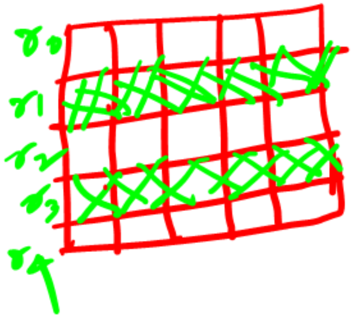
Show Empid and salary of all employee.

$$\Pi_{Empid,\,salary}(Employee)$$ → Relational Algebra.

Select Empid, salary from Employee; ⇒ SQL

Select(σ) :→

Select a particular row.



Show Empid, Empname, salary of all the employee whose salary > 10000.

$$\Pi_{Empid,\,Empname,\,salary}(\sigma_{salary>10000}(Employee))$$ → R.A

Select Empid, Empname, salary from Employee where salary > 10000; ⇒ SQL.

* Find the name of Employee and salary whose salary over 10000 and age is less than 40.

$$\pi_{Empname,\ salary}\left(\sigma_{salary > 10000 \text{ and } age < 40}(Employee)\right)$$

Select Empname, salary from Employee where salary > 10000 and age < 40;

Average() :→ Find the avg. salary of the employee's from the employee table.

$$\boxed{\Pi_{\mathcal{G}_{Avg(salary)}}(\text{Employee})}$$

* Find the name and salary of those employee who are withdrawing the salary grater than the avg. salary?

$$\boxed{\Pi_{Empname,\,salary}\left(\sigma_{sal > \mathcal{G}_{Avg(salary)}}(\text{Employee})\right)}$$

Rename ($\rho_x(E)$) :→ Returns the result of E under the name of x. The ρ_x operation used to rename the expressed temporary table.

Ex: 1

$$\rho_x \leftarrow \Pi_{salary}\left(\sigma_{salary < G_{max}(salary)}(Employee)\right)$$

$$\Pi_{Empname}\left(\sigma_{salary = G_{max}(x.salary)}(Employee \times x)\right).$$

name	salary
AA	10000
BB	120000
CC	9000
DD	7000
EE	25000

→ x

Ans. BB.

⇒ Find 2nd Highest salary of the employee.

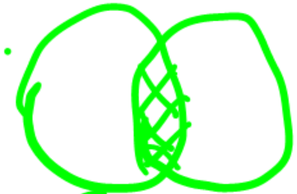
Union ($\cup$) :→ Borrower, Dipositer.

$$\Pi_{\text{cast-name}}(\text{Borrower}) \cup \Pi_{\text{cast-name}}(\text{Dipositer}).$$

B D.

Intersection ($\cap$) :→

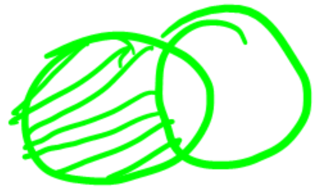
$$\Pi_{\text{castname}}(\text{Borrower}) \cap \Pi_{\text{castname}}(\text{Dipositer})$$



B D.

Set Difference ($-$) :→

$$\Pi_{\text{castname}}(\text{Borrower}) - \Pi_{\text{castname}}(\text{Dipositer}).$$



B D.

Cartesian Product :-> (X).

student

Rollno.	Name	DOB
1	AA	1/2/10
2	BB	1/3/12
3	CC	2/6/11

Marks.

Rollno	Total
1	353
3	448

step-1

student X Marks.

RollNo	Name	DOB	RollNo	Total
1	AA	1/2/10	1	353
1	AA	1/2/10	3	448
2	BB	1/3/12	1	353
2	BB	1/3/12	3	448
3	CC	2/6/11	1	353
3	CC	2/6/11	3	448

Spurious Tuple

step-2 :->

RollNo	Name	DOB	RollNo	Marks
1	AA	1/2/10	1	353
3	CC	2/6/11	3	448

step-3 :->

RollNo	Name	DOB	Marks
1	AA	1/2/10	353
3	CC	2/6/11	448

* Employee {Empname, Empno, salary, Age, Deptno}

Department {Deptno, Dname, Dloc, NOE}

Find the name of those employee who are works at 'Kolkata' and earning more than 15000.

$$\pi_{Empname}\left(\sigma_{Employee.salary > 15000 \land \sigma_{Employee.Deptno = Department.Deptno \land \sigma_{Department.Dloc = 'Kolkata'}}}(Employee \times Department)\right)$$

$\land \rightarrow$ and

$\lor \rightarrow$ or